

AgriCrop: Geospatial Plant Disease & Soil Moisture Intelligence Network

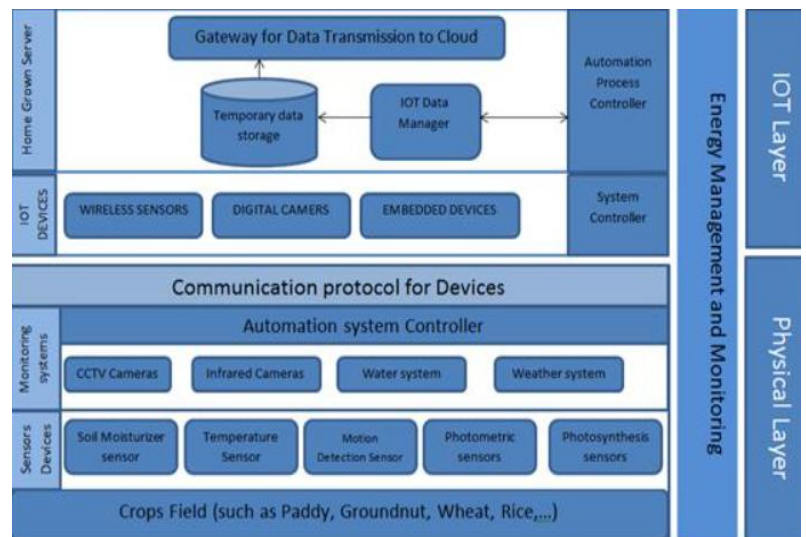
Submitted by:

K.Ribka

Ribka2974@gmail.com

1. Project Introduction

AgriCrop is an advanced AI-driven initiative designed to modernize precision agriculture. By leveraging geospatial data and real-time sensor analytics, the system provides an intelligent network capable of monitoring crop health and optimizing soil moisture management.



2. Project Objectives

- **Geospatial Analysis:** To utilize satellite and aerial imagery for large-scale crop health monitoring.
- **Disease Detection:** To implement computer vision models for the early identification of plant diseases.
- **Soil Moisture Intelligence:** To analyze soil data and provide actionable insights for efficient irrigation, reducing water wastage.



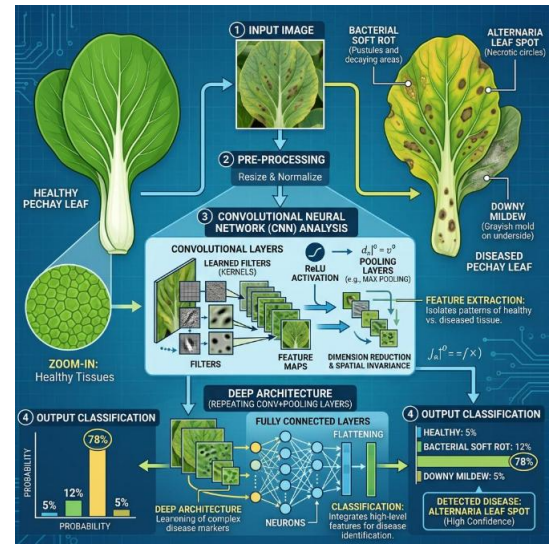
3. Technical Architecture

Component Technology Used

AI/ML Framework Python, TensorFlow, OpenCV

Data Processing Geospatial analysis libraries (GDAL/Rasterio)

Backend Integration API-driven architecture for real-time



monitoring

4. Implementation Strategy

5. Frontend Development: 1. Data Acquisition:

- Collect plant leaf images from the PlantVillage dataset.
- Collect soil moisture data using a soil moisture sensor or sample dataset.

2. Data Preprocessing:

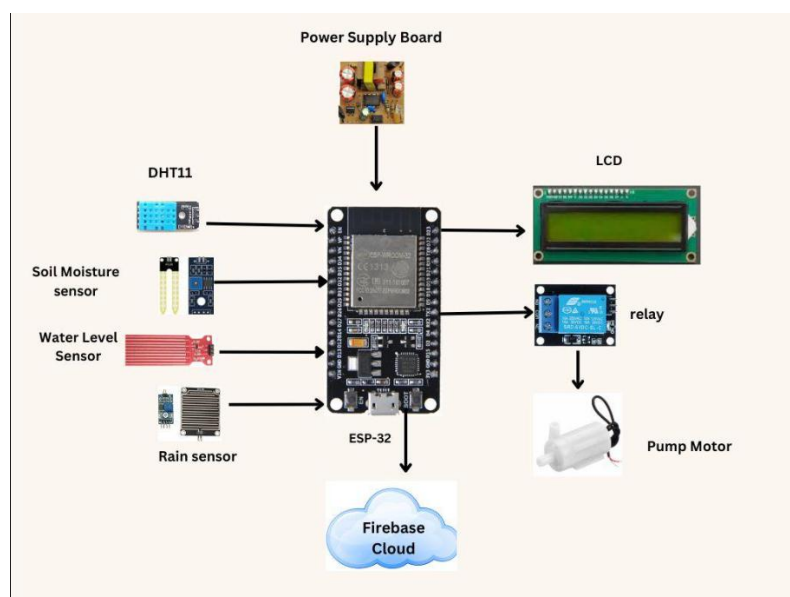
- Resize images, remove noise, and normalize data.
- Clean and prepare sensor data.

3. Model Training:

- Train a CNN model using TensorFlow for plant disease detection.
- Test the model and measure accuracy.

4. Backend Development:

- Build APIs using FastAPI to process image uploads and sensor data.



- Develop a simple dashboard using HTML/CSS/JavaScript or React.
- Display disease prediction, soil moisture level, and irrigation suggestions.

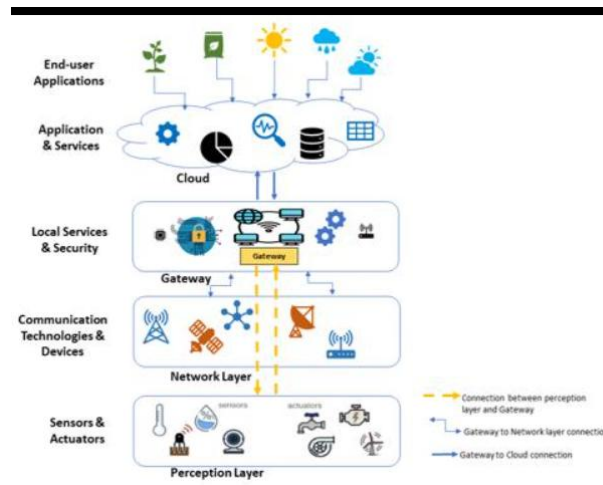
6. Deployment:

- Deploy the application locally or on the cloud and test the complete system.

5. Future Scope

- Drone-based crop monitoring.
- Satellite image integration.
- Weather prediction for irrigation planning.
- Mobile application for farmers.
- Multi-language support.

6. Conclusion



AgriCrop is an AI-powered smart agriculture system that detects plant diseases and monitors soil moisture. It helps farmers identify crop diseases early, optimize irrigation, reduce water wastage, and improve crop yield using Artificial Intelligence and Computer Vision. The AgriCrop project addresses critical challenges in modern agriculture by integrating AI into field operations, fostering sustainable farming practices through data-backed intelligence.